# Mirroring the Scene Flips the Score: Directional Success Depends on Layout in Language-Conditioned Manipulation

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***Abstract*—Language-conditioned manipulation has seen rapid progress, with generalist policies now following relational commands such as placing an object to the left or right of another. Directional success gaps between opposite commands have become a common way to assess whether these policies ground language, but such comparisons conflate several distinct questions, limiting a nuanced understanding of current capabilities.**

**In this paper, we test what a directional success gap measures. We separate the comparison into executed-action response, endpoint ordering, requested-side depth, and task completion. We then apply matched interventions in a controlled setting, holding the checkpoint, controller, and seeds fixed while changing either the object layout or the linguistic reference frame. We evaluate released checkpoints across two simulators with prespecified outcomes and equivalence tests. We find that matched layout interventions change the requested-side-depth contrast in every checkpoint that passes the prespecified endpoint-redirection control, and that layout alone can reverse which direction completes more often under an unchanged predicate. Moreover, in the reference-inversion cohort, restating the same physical goals from the other object's reference frame collapses the endpoint separation that familiar wording produces. These results show that directional success reflects both scene geometry and linguistic form, and that neither a large nor a small gap identifies grounding on its own. Project page: https://github.com/adeeb10abbas/steerable**

***Index Terms*—Vision-language-action models, robot learning, manipulation, language grounding, evaluation methodology.**

## I. Introduction

Vision-language-action (VLA) policies promise a single interface for open-ended robot control: a user states a goal in words, and one pretrained model produces the actions [1]–[4]. Whether that interface delivers is an empirical question, and relational placement has become a standard way to ask it. A representative protocol holds the scene fixed and asks the policy to place an object in "the pot on the left" or "the pot on the right." It balances trials between the two directions and scores each with a frozen success rubric [5]. It is tempting to read a large LEFT/RIGHT gap as a language-grounding failure: the model understood one direction but not the other, or vision overrode language [6].

That reading is plausible because robot datasets often use formulaic, low-diversity instruction labels, making the command predictable from the observation and allowing a conditioned policy to ignore language [7]–[9]. The attribution question is whether a binary directional score demonstrates that failure in a particular experiment [10].

That score combines several questions. For a trial to succeed, the instruction has to alter control, the change has to point toward the requested relation, the robot has to execute a feasible grasp and transport, and the final state has to cross a binary task boundary. A policy can pass the first two stages yet fail the last two because, for that checkpoint, one requested placement is harder to complete in the sampled configuration. The reverse mistake is also possible: a policy can react to the literal tokens LEFT and RIGHT without preserving the physical goal once the linguistic reference frame is inverted [11]. A single binary contrast can therefore produce two opposite errors: a false failure when language redirects the endpoint but scene-conditioned task competence differs by direction, and a false pass when familiar wording elicits the expected behavior without relation-level transfer.

Our central finding is an invariance failure: the directional gap is a property of the policy–layout pair, not a layout-invariant property of the checkpoint. Reflection reverses completion with checkpoint, prompts, controller, and seeds fixed (Fig. 1); Sec. IV gives the paired analysis.

Scene dependence is not the only threat to interpretation. When the same physical goals are expressed from the other object's reference frame—"cube left of bowl" becomes "bowl right of cube," and conversely—the familiar base-frame endpoint separation nearly vanishes. Base-frame sensitivity to LEFT and RIGHT therefore does not establish that the underlying relation transfers across linguistic frames.

The controls change what a directional score can support. A large gap can be a false grounding failure when one layout makes one requested placement harder to complete; a reassuring base-frame response can be a false pass when it does not survive the tested equivalent expression of the relation. Reflection exposes the first error, reference inversion exposes the second, and the endpoint decomposition makes both visible.

We make three contributions, ordered by the claims they support:

- **A matched object-layout intervention.** Moving only the movable objects, with the checkpoint, prompts, controller, and seeds unchanged, reverses which direction completes more often. Graded displacement and symmetric-scene controls show the same dependence in continuous placement.

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- **A reference-inversion control for relation-level transfer.** Holding each physical goal and the manipulated object fixed while restating the relation from the other object's frame removes the endpoint separation that the familiar wording produces.
- **A decomposition that separates redirection from completion.** Executed-action response, signed endpoint ordering, requested-side depth, and frozen task completion locate where a directional contrast changes, and expose layout-dependent placement even where binary success sits at ceiling.

None requires retraining, new hardware, or a change to the frozen success predicate.

## II. Related Work

**Generalist language-conditioned policies.** Modern manipulation policies pair pretrained visual-language representations with expressive action decoders. Representative systems include $\pi_0$ and $\pi_{0.5}$, which use flow-matching action models, OpenVLA, an open autoregressive alternative, and RT-2, which transfers web knowledge to control [1]–[4]. Hierarchical variants place a reasoning model above the controller and pass it natural-language subtasks [12]. Scale has expanded the tasks one policy can address, but it also makes a failed episode difficult to attribute: language interpretation, object localization, grasping, transport, release, and task scoring all lie on the same path to success.

**Steerability and instruction following.** A growing literature treats weak instruction following as the bottleneck and enriches the command interface. CAST augments datasets with counterfactual language [13], while STEER relabels demonstrations to expose a low-level control interface [14]. Steerable Policies train on subtask, motion, trace, and point commands [5], and InSight makes individual primitives addressable [15]. Other work steers a frozen policy at test time [16] or asks how steerability survives post-training [10], [17]. These methods address genuine failures of instruction following, but they rely on directional and relational success contrasts to certify that the failure was linguistic in the first place.

**Diagnosing whether language is used.** Counterfactual evaluation reduces attribution ambiguity by holding the visual input fixed while changing the instruction. LIBERO-CF and Counterfactual Action Guidance examine cases in which visual evidence dominates a conflicting instruction [6], and recent analyses bound how far instruction following transfers to unseen phrasings [11]. Steering protocols also report graded task progression, per-primitive progress, or success by perturbation condition [5], [15], [16]. Those approaches grade progress toward the command that was issued. Our contrast asks a different question: when two commands share the same reset and policy-sampling seed, what changes because one relation word changes? The exact pairing requires a resettable simulator, but it prevents scene and sampling differences from entering the prompt contrast [15].

**Evaluation practice.** Grounding spatial and directional language into executable behavior long predates VLAs, and early systems already separated the referring expression, the spatial relation, and the executed path [18], [19]. Recent large-scale evaluations, including the TRI study of Large Behavior Models, strengthen manipulation comparisons through controlled initial conditions, randomized trials, and statistical confidence [20]. That work makes comparisons between success rates more trustworthy. We ask what a directional success gap identifies: whether, under a frozen predicate, it can isolate language as the source of failure when the same gap also responds to where the objects sit.

## III. Experimental Design

We layer two controls on the same base-frame prompt pair. Reference inversion changes the linguistic frame while holding each physical goal fixed; the scene interventions change object geometry while preserving the prompt pair.

**Minimal-pair token intervention.** Literal prompt tokens appear in small capitals, LEFT and RIGHT, while *leftward* and *rightward* denote physical directions in robot coordinates; a policy can track the token without preserving the direction it denotes. For each reset $s_i$ we run a policy under prompts $\ell_i^L$ and $\ell_i^R$ that differ only in one opposite direction word, holding scene state, checkpoint, controller, environment seed, policy-sampling seed, action interface, and horizon fixed. Both prompts therefore execute from the same initial state and random draws; the direction word is the designed difference. We issue the instruction once at reset and leave it alone: no oracle actions, no prompt switching, and no language conditioned on progress. DROID/RoboLab pairs the DROID Franka embodiment [21] with the RoboLab Isaac Sim environment [22] and uses a Rubik's cube and bowl; RoboTwin [23] uses a wooden block and playing-card box. Both require placement and release in the requested relational region.

**Checkpoints and pre-intervention screen.** Table I lists the eight expanded Phase-A rows; App. D records the available artifact identifier and revision behind each one. Two are action-only VLAs and six are world-action models (WAM), which predict future observations alongside actions. Within each arena, checkpoints use its fixed action interface and frozen predicate. Edge and Nano are the 4B and 16B policies of one Cosmos3 family [24], so the rows are not fully independent. The checkpoints come from different developers with different pretraining mixtures and post-training recipes; exact training-episode membership, caption exposure, and sampling weights are undisclosed. Every causal contrast is therefore within checkpoint. Cross-row differences provide context only; no comparison across rows is load-bearing.

We define the pre-intervention screen as one completed matched-pair row per checkpoint and arena. Repository evidence spans eleven checkpoint identities. Of these, the Phase-A screen reported here comprises eight checkpoint–arena rows: five DROID/RoboLab rows at 27 matched pairs each and three RoboTwin rows at 63 matched pairs each, for 324

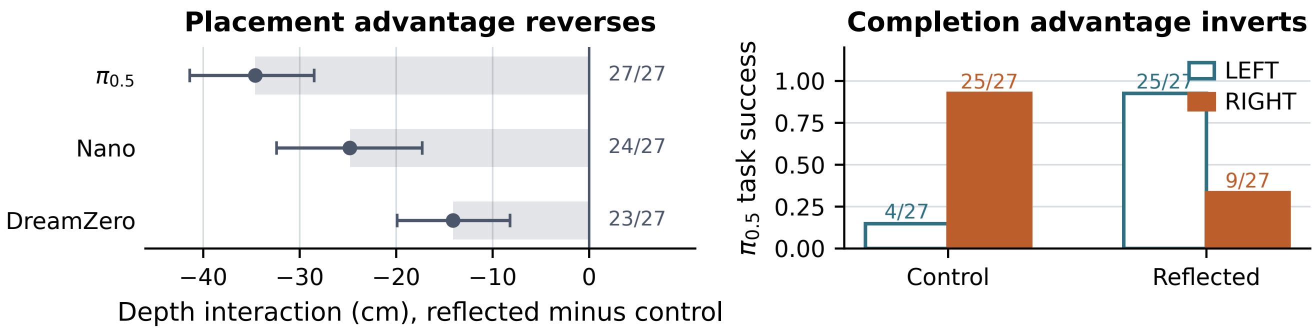


Fig. 1. **Reflection changes continuous placement and reverses $\pi_{0.5}$ completion.** Left: reflected-minus-control interaction in requested-side depth, with seed-clustered 95% bootstrap intervals; fractions give seeds with a negative interaction. Right: the $\pi_{0.5}$ completion advantage reverses after reflection; plotted fractions give the success counts. All causal comparisons are within checkpoint; the panels do not support cross-model ranking.

TABLE I
EXPANDED PHASE-A CHECKPOINT–ARENA ROWS IN THE PRE-INTERVENTION SCREEN; SHORT NAMES IN PARENTHESES.

| Model (short name) | Class | Pairs |
|---|---|---|
| *DROID/RoboLab: cube to the requested side of a bowl* | | |
| $\pi_{0.5}$ DROID [4] | VLA | 27 |
| GR00T N1.7 DROID (GR00T) [25] | VLA | 27 |
| Cosmos3 Edge Policy DROID (Edge) [24] | WAM | 27 |
| Cosmos3 Nano Policy DROID (Nano) [24] | WAM | 27 |
| DreamZero DROID (DreamZero) [26] | WAM | 27 |
| *RoboTwin: block to the requested side of a card box* | | |
| Efficient-WAM-RT [27] | WAM | 63 |
| FastWAM [28] | WAM | 63 |
| LingBot-VA [29] | WAM | 63 |

pairs or 648 episodes. The remaining identities appear only in separate cohorts: a 20-pair $\pi_0$-FAST compatibility bridge and two bounded three-pair RoboTwin gates, LingBot-VLA 4B (distinct from the screened LingBot-VA checkpoint) and Light-WAM. Because those cohorts use different designs and denominators, we do not pool them into the Phase-A screen. The two arenas use different tasks and different success tests, so no outcome estimate or effect pools arenas; totals across rows are bookkeeping only, and the larger intervention cohorts state their own denominators.

**Behavioral measurements.** Let $a^L_{1:T_L}$ and $a^R_{1:T_R}$ be the executed action sequences in a minimal pair, and let $y^L_i, y^R_i$ be the final target-minus-reference lateral offsets in robot coordinates, where positive is robot-left. The four measurements correspond to four stages of the task: prompt response, directional ordering, placement quality, and completion. We report

$$A_i = \mathbb{1}\left[a^L_{1:T} \neq a^R_{1:T}\right], \qquad T = \min(T_L, T_R), \tag{1}$$

$$D_i = y^L_i - y^R_i, \tag{2}$$

$$S^d_i = \mathbb{1}[\text{frozen task predicate is satisfied for } d], \tag{3}$$

$$M^L_i = y^L_i, \qquad M^R_i = -y^R_i. \tag{4}$$

$A_i$ uses bitwise inequality on the complete common executed prefix. Positive $D_i$ means the requested endpoint ordering: within a pair the object came to rest further leftward under LEFT than under RIGHT. $S^d_i$ is task completion, and requested-side depth $M^d_i$ records where the target comes to rest along the requested physical direction.

**Calibrating $A_i$ against sampling noise.** $A_i$ tests response, not semantics, and alone it is weak evidence: stochastic sampling makes two executions differ even under an unchanged prompt. We therefore query each policy on one frozen observation, without executing the robot, to separate prompt response from sampling noise; App. B gives the construction and full results.

**Reference inversion.** Base-frame prompts change the surface token and physical goal together; the reference-inversion control instead holds the physical relation fixed and changes its linguistic frame. The base-frame prompts are "Put the Rubik's cube to the left of the bowl." and "Put the Rubik's cube to the right of the bowl." The inverted prompts are "Place the Rubik's cube so that the bowl is to the right of the Rubik's cube." and the same sentence with *left*. All four name the cube as the object to place, so inversion changes the frame of the relational clause and not the manipulandum; the direction word flips while the intended physical relation does not. The scorer, and the sign of $M$ for this control, follow the physical goal rather than the prompt text. We run 341 resets for the $\pi_{0.5}$ study in one symmetric DROID/RoboLab scene, with both forms of both physical goals sharing reset and sampling state, for 1,364 episodes on seeds disjoint from every scene-intervention cohort.

**Geometry interventions.** Reflection is a pure world-space object-position intervention fixed in advance: it mirrors the centers of all movable objects about the robot sagittal plane without mirroring observations or un-mirroring actions. The robot, cameras, fixed scene geometry, prompts, seeds, and controller remain unchanged. We pair it with two further interventions fixed in advance: a seven-level sweep that moves only the reference object laterally and a symmetric-scene cohort that drives a registered movable-object asymmetry metric to zero (App. A). The symmetric package also changes which other objects are present, so it is not a position-only intervention and does not imply bilateral embodiment symmetry. We interpret scene effects only for rows that pass the prespecified endpoint-redirection positive control, which asks whether the policy still moves its endpoint in the commanded direction. We report failed rows without a geometry interpretation.

**Scoring and inference.** We do not relax the original task predicate. Each episode receives exactly one outcome label, assigned in a fixed order: pick failed, transport failed, wrong side, release failed, or correct. Every behavioral failure stays in its denominator. Marginal binary rates use Wilson intervals, paired continuous contrasts use seed-clustered percentile bootstraps with 20,000 resamples and exact sign tests, and layout interactions use exact within-seed label permutations. Absence claims require two one-sided equivalence tests (TOST) against prespecified margins, rather than nonsignificance [30]–[32]. Prompts, seeds, success tests, target quantities, margins, and the failure-label order are all fixed before the runs they affect. The public archive[1] records the available manifests, seeds, checkpoint revisions, and freeze times for the reported cohorts.

**Every base-frame pair changes the executed action, and most redirect the endpoint correctly:** All 324 minimal pairs across the eight screen rows execute different action sequences: 135 DROID/RoboLab pairs and 189 RoboTwin pairs. Endpoint ordering is positive in $125/135$ and $128/189$ pairs, respectively. These differences serve as the matched trace-difference positive control; they do not show that the policy represents the relation independently of its wording or guarantee task completion.

**Open-loop action response is not a standalone grounding score:** On a frozen observation, $\pi_{0.5}$ shows small but consistently oriented prompt differences, whereas DreamZero's prompt-to-noise ratio is undefined because its measured cross-seed variation is zero. These action-only responses are a positive control, not a standalone grounding score; App. B reports the diagnostic details.

## IV. Reflection Reverses Directional Completion

**Reflection reverses which direction $\pi_{0.5}$ completes:** The paired outcomes are nearly mirrored: control RIGHT and reflected LEFT each succeed in $25/27$ trials, whereas control LEFT and reflected RIGHT succeed in $4/27$ and $9/27$. The latter five-success difference is the part that reflection does not remove.

The eight-row Phase-A screen is descriptive context (Fig. 2). $\pi_{0.5}$ and DreamZero favor RIGHT, Nano is near ceiling, and GR00T completes only $3/27$ LEFT and $0/27$ RIGHT trials; $49/54$ GR00T episodes fail at pickup. The RoboTwin screening gaps are descriptively small. Two checkpoints later fail the endpoint-redirection positive control, so those rows do not test layout dependence.

### A. Pure object-position reflection

We reflect the movable-object centers about the sagittal plane (Sec. III) and evaluate four conditions per checkpoint (two directions by two layouts) over 27 seeds, or 108 episodes for each of three DROID/RoboLab checkpoints. The $\pi_{0.5}$ reflection cohort uses seeds disjoint from the screen. Figure 2 therefore reports $5/27$ and $24/27$, whereas the control condition below has $4/27$ and $25/27$. The primary outcome is the within-seed layout interaction in requested-side depth; success is secondary because binary outcomes can saturate by layout and direction. Nano, the checkpoint closest to the binary ceiling, illustrates why in Fig. 3.

**Reflection shifts the requested-side-depth contrast in the same direction in all three tested checkpoints:** The reflected-minus-control interaction is negative in every reflection checkpoint (Fig. 1). The interaction is $-34.6\,\mathrm{cm}$ for $\pi_{0.5}$ (95% CI $[-41.4, -28.5]$; 27/27 seeds; $p = 1.49 \times 10^{-8}$). It is $-24.8\,\mathrm{cm}$ for Nano ($[-32.4, -17.3]$; $24/27$; $p = 4.92 \times 10^{-5}$) and $-14.1\,\mathrm{cm}$ for DreamZero ($[-19.9, -8.2]$; $23/27$; $p = 3.11 \times 10^{-4}$).

**Placement shifts much more than endpoint redirection for $\pi_{0.5}$ and Nano:** Against those requested-side-depth shifts, endpoint redirection changes by $-1.12\,\mathrm{cm}$ for $\pi_{0.5}$ and $+0.5\,\mathrm{cm}$ for Nano. The two redirection interactions each have $p = .701$ because their signs split 12/15 and 15/12 over the same 27 seeds. We cannot call either redirection interaction unchanged; the equivalence test does not support that claim.

**The completion advantage flips under the unchanged frozen predicate:** Each seed contributes $(S_i^{R,\mathrm{refl}} - S_i^{L,\mathrm{refl}}) - (S_i^{R,\mathrm{ctrl}} - S_i^{L,\mathrm{ctrl}})$, so a negative value means reflection reduces or reverses the control-layout RIGHT completion advantage. In this $\pi_{0.5}$ DROID/RoboLab cohort, the mean per-seed interaction is $-1.37$ on its $[-2, 2]$ scale ($p = 5.96 \times 10^{-8}$). The control-layout gap is not an artifact of one policy draw: eight samples in each of 27 fixed scenes yield $41/216$ LEFT and $197/216$ RIGHT successes. This resampling cohort covers the control layout only.

**A fixed directional marginal cannot explain the reversal by itself:** The checkpoint is unchanged across layouts, so a constant exposure-driven preference predicts a stable preference rather than a reversal. Exposure conditioned on object configuration remains consistent with the result. These experiments do not distinguish a physical effect of geometry from coverage correlated with object configuration. Section VIII states the experiment needed to separate them.

### B. A graded intervention and converging controls

**The dependence is graded across the seven tested positions:** We move the bowl across seven lateral positions while retaining the same Nano checkpoint and 15 matched seeds per level (210 episodes). Across all seven registered positions, the requested-depth contrast has a slope of 1.12 cm of depth per cm of reference displacement (95% CI $[0.72, 1.56]$); 13 of 15 seed-level slopes are positive (exact $p = .00739$). A post hoc analysis of the middle five positions gives a weaker slope: 0.49 (95% CI $[-0.04, 0.97]$), with 11 of 15 positive slopes ($p = .118$). We therefore limit the graded-dependence claim to the full registered range. The middle-five analysis does not support an interior linear law. The fitted zero crossing lies outside the positions we tested, so we cannot locate a balanced interior layout from these data.

[1]https://github.com/adeeb10abbas/steerable

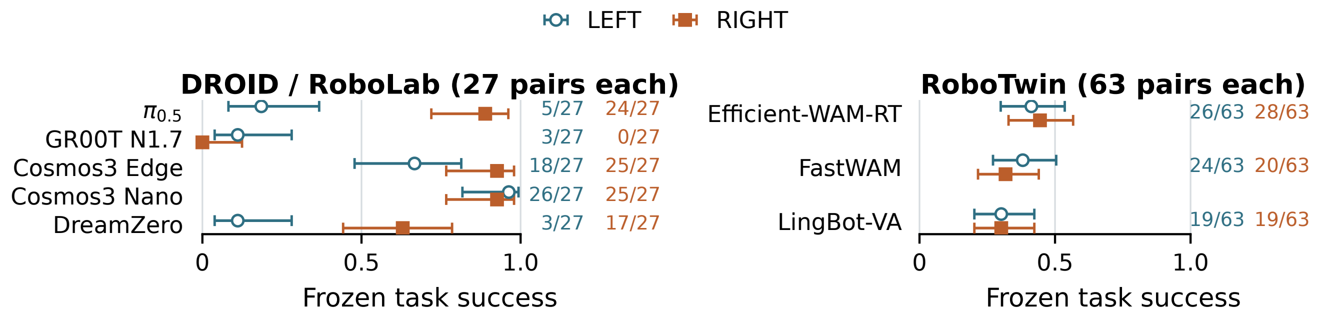


Fig. 2. **Directional task success varies across the eight checkpoint–arena rows.** Points show marginal LEFT and RIGHT success with 95% Wilson intervals. DROID/RoboLab uses 27 matched pairs per row and RoboTwin 63; their tasks and frozen predicates differ. The screen is descriptive, and every causal comparison in this paper is within checkpoint; the figure does not support cross-model or cross-arena ranking.

**Three registered Nano interventions change directional placement or completion:** The seven-position sweep shows why binary success alone is a weak diagnostic: LEFT completion changes from $15/15$ to $10/15$, while RIGHT stays close to ceiling (Fig. 4). Separate registered Nano cohorts support the same layout-dependence claim without being pooled with reflection. For Nano, the RIGHT-minus-LEFT success gap changes from $(22-26)/27 = -4/27$ when the target starts robot-left of the reference to $(27-23)/27 = +4/27$ when it starts robot-right. Under the registered right-start-minus-left-start convention, the interaction is therefore $+8/27 = +0.296$ ($p = .0156$). Also for Nano, swapping target and reference roles changes the depth contrast by $+13\,\mathrm{cm}$ (95% CI $[6, 21]$, $p = .00151$). The role swap keeps the direction word and relation template fixed, but changes the manipulandum and physical goal. Reference inversion instead keeps both fixed and changes the linguistic frame.

## V. Reference Inversion Collapses Endpoint Separation

**Reference inversion collapses endpoint separation on matched physical goals:** For $\pi_{0.5}$, reference inversion sharply reduces the observed familiar-frame separation when the same physical relations are expressed from the other object's perspective (Fig. 5).

Reference inversion tests whether the same physical relation survives being expressed from the other object's frame. Under inversion the direction word and the physical goal point opposite ways, which separates two behaviors that base-frame prompts cannot tell apart. If the policy tracks the relation, endpoint position for the leftward goal minus that for the rightward goal should be positive. If it tracks the direction word regardless of the sentence frame, the contrast should be negative.

**The inverted response matches neither relation transfer nor token mirroring:** Base-frame prompts separate the endpoints by $+23.28\,\mathrm{cm}$ (95% CI $[21.89, 24.56]$), whereas reference-inverted prompts yield only $+0.19\,\mathrm{cm}$ ($[-0.88, 1.27]$). The observed sign and magnitude support neither relation-preserving transfer nor simple token mirroring, which would reach $-23.28\,\mathrm{cm}$ if the base-frame response transferred unchanged. The interval includes zero, but the equivalence test does not establish equivalence to zero. The experiment does not distinguish failed relation transfer from sensitivity to out-of-distribution phrasing. This conclusion concerns endpoint direction only. All 682 base-frame/inverted physical-goal pairs execute different action traces.

**Inversion also reduces placement quality for both physical goals:** The inverted-minus-base-frame change is $-9.78\,\mathrm{cm}$ for the leftward goal (90% CI $[-10.92, -8.59]$). For the rightward goal, it is $-13.31\,\mathrm{cm}$ ($[-14.07, -12.54]$). Both intervals lie outside the prespecified $\pm 4.15\,\mathrm{cm}$ equivalence margin. Success falls from $210/341$ to $151/341$ for the leftward goal and from $297/341$ to $240/341$ for the rightward goal. These successes make a wholesale switch to moving only the bowl less consistent with the evidence because the predicate scores placement of the cube. Because reference-object displacement was not measured, the counts do not exclude incidental bowl motion or a mixed strategy. The corresponding changes, $-17.30$ and $-16.72$ percentage points, do not meet the prespecified $\pm 15.56$-point equivalence criterion. Base-frame response is therefore insufficient evidence that a physical relation is preserved across reference frames.

## VI. Binary Ceilings Hide Placement Change

**A zero binary interaction can hide a large placement change:** Nano's completion interaction is zero ($p = 1$), whereas requested-side depth shifts by $-13.9\,\mathrm{cm}$ ($p = 1.8 \times 10^{-4}$).

The symmetric-scene cohort contains 4,096 episodes (2,048 matched pairs) across five checkpoint–arena rows. The dense rows contribute 763 $\pi_{0.5}$ pairs and 1,123 Nano pairs; Edge, DreamZero, and FastWAM contribute 54 each. The intervention zeros the registered movable-object asymmetry metric but replaces a scene package rather than changing positions alone. Four DROID/RoboLab checkpoints pass the endpoint-redirection control and support a scene-effect interpretation. FastWAM and the separate 108-episode LingBot-VA cohort fail it; we report both without a geometry interpretation. GR00T is excluded because it almost never completes. The $\pi_{0.5}$ seeds are disjoint from those used for reference inversion.

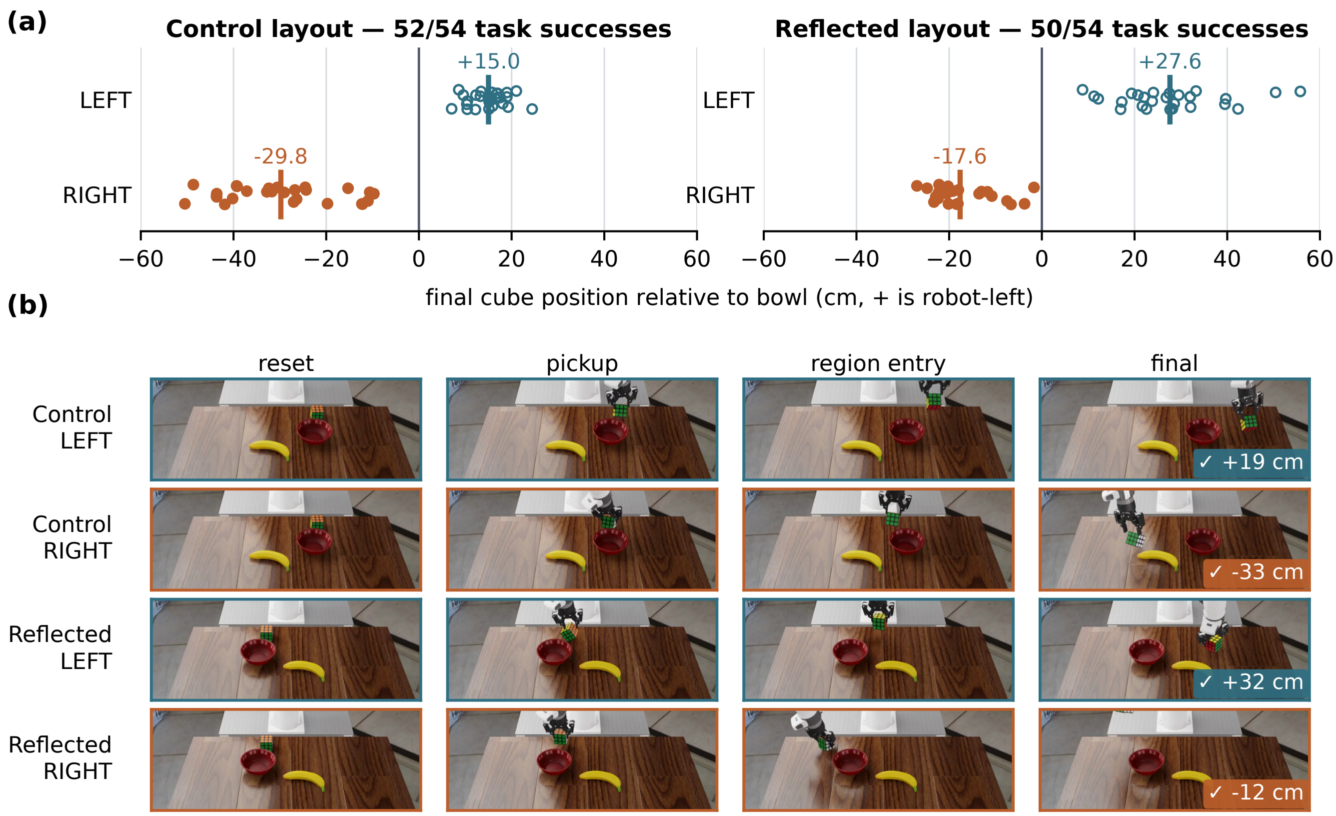


Fig. 3. **Reflecting movable-object positions moves the placement while the binary score stays at ceiling (Nano, 108 episodes).** (a) Final lateral cube position for every seed in the four conditions; bars mark their means and titles give task completion. The prompts still separate after reflection, by 45.3 cm against 44.8 cm in the control layout, but both distributions shift by about 12 cm, producing the requested-side-depth interaction reported below. Completion moves only from 52/54 to 50/54. (b) The executed rollout for the lowest released seed in each condition, sampled at that episode's logged events, with ticks marking the frozen predicate and numbers repeating the final signed offset. Robot-left appears toward the image right in this viewport, so the signed values rather than the images fix direction. Rollouts were selected by lowest released seed before any outcome was read.

**The symmetric scene shrinks the $\pi_{0.5}$ gap without meeting equivalence:** For $\pi_{0.5}$, 341 prompt pairs at each endpoint show the binary RIGHT-minus-LEFT gap falling from 74.2 to 20.2 percentage points and the requested-depth gap from 17.0 to 5.75 cm. At the symmetric endpoint, the paired 90% interval for the binary gap is $[15.0, 25.2]$ points against a prespecified $\pm 15.56$-point margin. The requested-depth interval is $[4.85, 6.66]$ cm against $\pm 4.15$ cm. Neither meets its equivalence criterion. The prespecified power analysis at $n = 341$ gives an 80%-power minimum detectable effect of 7.8 percentage points for the binary gap, relative to the 15.56-point equivalence margin. We also analyze a separately registered matched core of 27 seeds shared across layouts. Its binary and requested-depth interactions are $-51.9$ points (95% CI $[-81.5, -22.2]$, $p = .00786$) and $-12.4$ cm (95% CI $[-19.7, -4.9]$, $p = .00348$), respectively. Base-frame redirection at the symmetric endpoint is $+23.9$ cm, with 335/341 positive pairs.

**DreamZero's binary contrast and Edge's depth contrast change under the scene intervention:** The other positive-control-passing checkpoints provide two further within-checkpoint checks (Fig. 4). DreamZero's observed binary gap changes from $+51.9$ to $-3.7$ points (interaction $-55.6$ points, $p = .00085$), bringing the measured gap near zero without establishing equivalence. Edge's $-14.8$-point binary interaction is inconclusive: its 95% interval $[-40.7, +7.4]$ includes zero. Its depth interaction is $-19.2$ cm (95% CI $[-26.0, -11.5]$).

Appendix C reports the descriptive failure-label bookkeeping for the full $\pi_{0.5}$ endpoint cohorts.

## VII. Discussion

**A binary score can fail in opposite directions.** Reflection shows how layout-dependent completion can be misread as a grounding failure; reference inversion shows how a reassuring familiar-frame response can be misread as relation-level grounding. The measurements locate both errors: a binary ceiling can hide a placement change, and an executed trace can change without expressing the intended relation. The frozen predicate remains the final completion test, but it cannot identify the cause.

**Binary thresholding can expose or hide placement shifts.** The frozen predicate maps pickup, transport, release, and placement into one completion label. A layout-dependent endpoint shift can coincide with a reversed completion advantage, as under $\pi_{0.5}$ reflection, or with no change in the binary

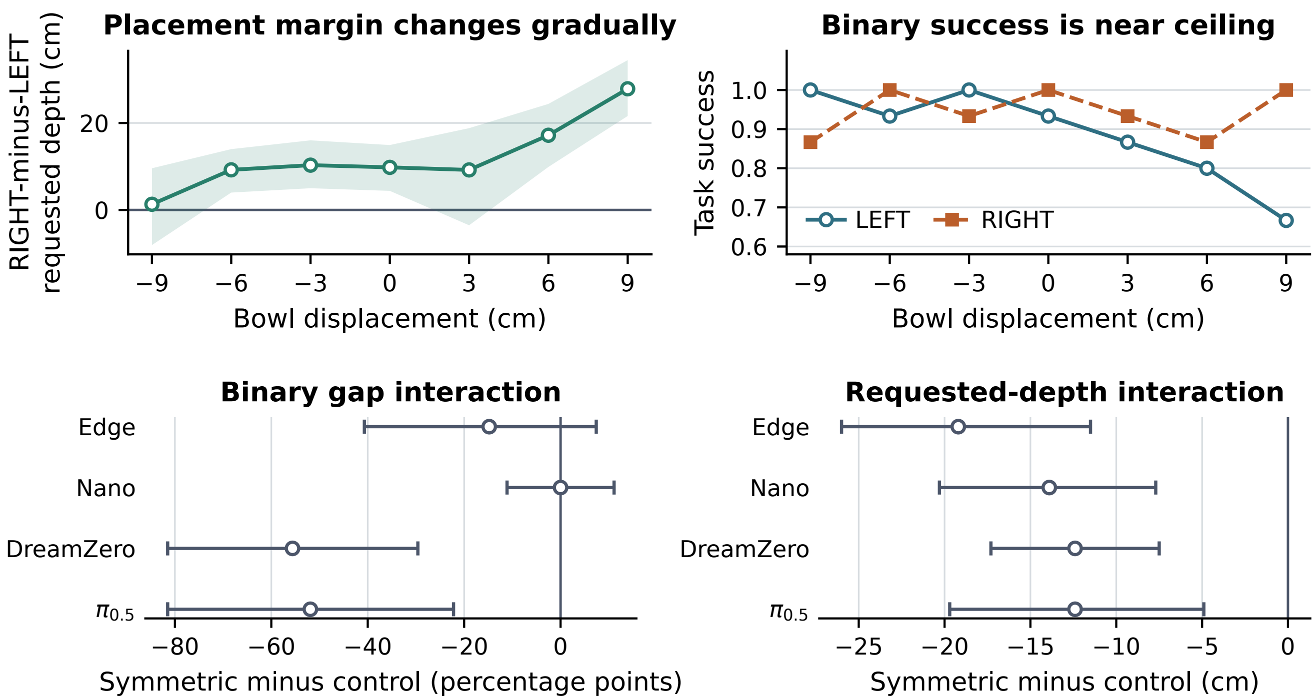


Fig. 4. **Two scene interventions, measured continuously and binarily.** Top: for Nano, the directional placement contrast across the seven-position reference-object sweep, with mean RIGHT-minus-LEFT requested-side depth and seed-clustered 95% intervals on the left and frozen task success by requested direction on the right; each position contains 15 matched seeds. Bottom: symmetric-minus-control interactions in the matched 27-seed core of the positive-control-passing DROID/RoboLab cohort, with seed-clustered 95% intervals; requested-depth disparity decreases in every checkpoint. In both interventions the continuous measure exposes variation that binary success compresses near ceiling, as for Nano. Every comparison is within checkpoint; the panels do not support cross-model ranking.

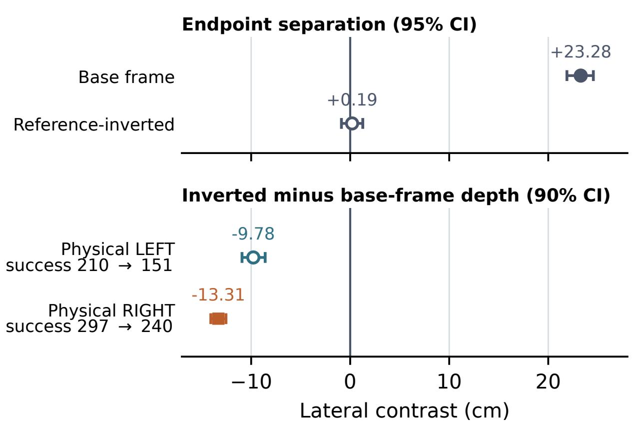


Fig. 5. **Reference inversion collapses directional endpoint separation for $\pi_{0.5}$.** Top: leftward-minus-rightward physical-goal endpoint separation with seed-clustered 95% bootstrap intervals; positive is relation-consistent. Bottom: inverted-minus-base-frame requested-side depth with 90% intervals; labels give base-frame-to-inverted success counts out of 341. This cohort's seeds are disjoint from those of the scene-intervention experiments.

interaction, as at Nano's ceiling. These cohorts do not establish which task stage or boundary crossing produces the difference. They show that completion can reveal or conceal a continuous placement response without identifying its source.

**A directional gap belongs to the policy–layout pair.** For $\pi_{0.5}$, reflection changes requested-side depth by $-34.6$ cm while the policy, prompts, and seeds remain fixed. The tested symmetric scene shrinks the binary gap but leaves a 20.2-percentage-point residual. The controls therefore show that the score depends on layout; they do not identify what produces the remainder.

**What a directional evaluation needs in order to be interpretable.** For relational placement, we recommend matched base-frame LEFT/RIGHT prompts under identical reset and policy seeds, paired with action distinctness, a signed endpoint measure, requested-side depth, and the frozen success predicate. Where stochasticity is material, add same-prompt repeats. Attribution further requires at least one matched scene intervention with a preserved endpoint-redirection positive control and a reference inversion that preserves the physical goal. Claims that an effect disappeared should rest on equivalence tests rather than a nonsignificant difference.

**Directional gaps require matched scene controls.** Directional and relational success contrasts are often used to support a linguistic attribution. Our results show that, in this evaluation, a directional gap and a layout effect are not separable without a matched scene intervention. A published directional gap measured without that control may therefore reflect the evaluation scene.

## VIII. Limitations

Our design trades external validity for control. Exact state matching needs a resettable simulator, so hardware contact uncertainty and natural variation in human instructions remain untested. We interpret object-layout effects only for the DROID/RoboLab rows that pass the endpoint-redirection control; the RoboTwin rows screen behavior, not a cross-arena replication or refutation. The task family is narrow as well: two named objects, one lateral relation, and one reference inversion on one $\pi_{0.5}$ checkpoint. DreamZero's public demonstrations include lateral placements relative to a bowl, so this task may be closer to its demonstrated set than the other rows'.

The interventions establish scene dependence but do not distinguish physical reachability from training coverage correlated with similar object configurations. As Sec. IV explains, reflection rules out a fixed directional marginal as the complete explanation; configuration-conditioned exposure remains consistent with the result. The evaluated releases disclose neither episode membership, sampling weights, training layouts, nor configuration-conditioned captions, and corpus-wide LEFT/RIGHT counts would not recover the exposure relevant to these scenes. That attribution is therefore not resolvable for these released checkpoints. Two measurement gaps remain. We did not track reference-object displacement under inversion, although inverted success rates show that the cube still satisfied the placement predicate (Sec. V). The fixed-observation diagnostic measures action space only. Settling the exposure question requires training policies from scratch on corpora with controlled LEFT/RIGHT balance within matched object configurations and then rerunning the same interventions. Released checkpoints cannot provide that test.

## IX. Conclusion

Directional success does not isolate language grounding. Matched reflection and reference inversion show that layout and linguistic frame change the score, while Nano shows why completion alone can miss placement change. Directional grounding therefore requires controls that separate language response from scene-conditioned competence.

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## APPENDIX A
## SYMMETRIC-SCENE ASYMMETRY METRIC

The registered metric is the root-sum-square departure of the movable objects from bilateral symmetry about the robot midline. It measures midline objects from the midline itself and mirror-paired objects from each other's reflection, with position weighted at 10 per metre and orientation at 1 per radian. The control layout scores 3.85 and the symmetric layout 0.00, within prespecified millimetre and half-degree tolerances. The robot, cameras, wrist mounting, reset posture, prompts, seeds, and controller are unchanged across the two scenes, but the companion-object inventory changes; the metric therefore certifies movable-object symmetry rather than full-scene or embodiment symmetry.

## APPENDIX B
## FIXED-OBSERVATION ACTION DIAGNOSTIC

We issue 336 action requests across six checkpoint–layout conditions: three DROID/RoboLab checkpoints crossed with the control and reflected scenes. Each condition contains 27 matched LEFT/RIGHT seed pairs and one exact repeat per direction at seed 9400; all 12 repeat comparisons are bit-identical. For seed $s$, prompt effect is $\mathrm{RMS}(a_s^L, a_s^R)$ over the complete native returned action arrays. The noise pool contains $\mathrm{RMS}(a_s^d, a_{s'}^d)$ for each prompt $d$ and every distinct-seed pair $s < s'$. The prompt-to-noise ratio divides the median of the 27 prompt effects by the median of the 702 pooled same-prompt distances. The exact repeats verify reproducibility and do not define the denominator; when the pooled cross-seed median is zero, the ratio is undefined and we report the prompt-effect numerator alone.

For $\pi_{0.5}$, the prompt-induced median action difference is $4.1\%$ of the pooled same-prompt cross-seed median in control and $3.3\%$ after reflection, while its difference vectors share a consistent orientation across seeds (mean cosine $0.540$ and $0.652$; permutation $p < 10^{-5}$). For Nano, the corresponding ratios are $0.720$ and $1.192$, but the orientations are less consistent (mean cosine $0.027$ and $0.079$; permutation $p = .167$ and $.00314$). For DreamZero, all 702 same-prompt cross-seed distances are zero in each layout, while median prompt RMS is $0.00923$ in control and $0.02348$ after reflection. Its ratio is undefined rather than small; whether the zero measured variation belongs to the checkpoint or to harness seeding remains unresolved. No returned action in this diagnostic is executed, so these results support an action-response positive control rather than a task-success claim.

## APPENDIX C
## SYMMETRIC-SCENE FAILURE LABELS

The full $\pi_{0.5}$ endpoint cohorts pool directions within 682 episodes per layout. These unpaired totals are descriptive bookkeeping, not a separate interaction test.

| Layout | Pick | Transport | Wrong side | Release | Total |
|---|---|---|---|---|---|
| Control | 1 | 267 | 41 | 2 | 311 |
| Symmetric | 0 | 158 | 7 | 0 | 165 |

## APPENDIX D
## EVALUATED CHECKPOINT PROVENANCE

Each row below lists the recorded artifact identifier and revision used in the evaluation. The public manifests provide the available content hashes and runtime metadata. They do not expose training-episode membership, sampling weights, or configuration-conditioned caption exposure.

| Short name | Released artifact identifier and revision |
|---|---|
| $\pi_{0.5}$ | `pi05_droid_jointpos_polaris`<br>`v2a010-manifest-f5a56d9565f9381cc` |
| GR00T | `nvidia/GR00T-N1.7-DROID`<br>`05e7cc97e40dbd33b0890c35cc0214fcb0547ab5` |
| Edge | `nvidia/Cosmos3-Edge-Policy-DROID`<br>`3ea407af3e156c0af3b4bb6edd85842cc9a58777` |
| Nano | `nvidia/Cosmos3-Nano-Policy-DROID`<br>`6706d7680581c255ff61e0f3bb49d90eac55c79e` |
| DreamZero | `GEAR-Dreams/DreamZero-DROID`<br>`96ad344138c66e82536422432ad742f015784942` |
| Efficient-WAM-RT | `jiajun0613/Efficient-WAM_RoboTwin/`<br>`Efficient-WAM-RT`<br>`81280a79e8ac69dd6ffb9ce8698e00d122ec07fd` |
| FastWAM | `yuanty/fastwam/`<br>`robotwin_uncond_3cam_384.pt`<br>`139eebb6d90cdd9bdbbe465f72c6edc9ad5a518a` |
| LingBot-VA | `lerobot/lingbot_va_robotwin`<br>`d1e1f93a84eaf9bca9880856fda800cc98cc8eaa`<br>`robbyant/lingbot-va-posttrain-robotwin`<br>`8c9dea8abbc5c91cc9e18bc3264b8915083bbe70` |

The $\pi_{0.5}$ row is an internal release rather than a public repository revision, so its recorded identifier is a manifest digest, abbreviated here and given in full in the released manifest. LingBot-VA is evaluated as a base checkpoint plus a RoboTwin post-training checkpoint, so both revisions are required to reproduce it.